DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (A)
Osmania University, Hyderabad-07

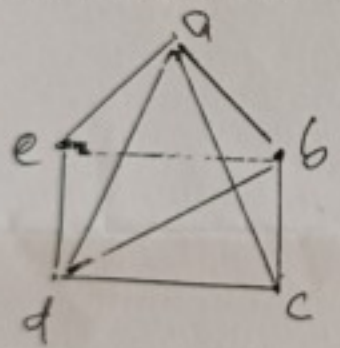
CLASS TEST-II

100523729008

Subject : Graph Theory

Class: B.E.(AI & ML)
Semester: IV
Academic Year: 2024-2025

Max Marks: 30
Duration: 1 1/2(2:30 p.m. to 4:00 p.m.)
Date : 27-06-2025

	PART A Answer All Questions (5 x 2 = 10 M)	Marks	BT	CO
1. a)	Explain Hopcraft – Karp algorithm for finding maximum matchings in bipartite graph.	2	2	3
b)	Define following : a) Independent sets b) Coverings	2	1	2
c)	State and Prove Eluler's formula in case of tree.	2	3	3
d)	Define line-connectivity and point-connectivity and Prove $\kappa(G) \leq \lambda(G) \leq \delta(G)$	2	2	2
e)	Is the following graph is planar or not . If planar re-draw it. 	2	2	2
	PART B **Answer Any Two Questions (2 x 10 = 20 M)**			
2.	List basic rules for construction of Hamiltonian paths or cycles. OR State and Prove Grinberg Theorem	10	3	3
3.	Write greedy and approximate algorithm for set cover.	10	2	3
4.	Prove that : A connected graph is Isomorphic to its line graph if only if it is a cycle.	10	3	3